

Total No. of Questions : 6]

SEAT No. :

P23

[Total No. of Pages : 2

APR -17/ BE/ Insem. -25

B.E. (Mechanical)

DESIGN OF PUMPS, BLOWERS AND COMPRESSORS (Elective - IV)

(2012 Pattern) (Semester - II)

Time : 1 Hour]

[Max. Marks : 30

Instructions to the candidates:

- 1) *Solve Q. 1 or Q.2, Q.3 or Q.4 and Q.5 or Q. 6.*
- 2) *Figures to the right indicate full marks.*
- 3) *Neat diagrams must be drawn, wherever necessary.*
- 4) *Use of electronic pocket calculator is allowed.*
- 5) *Assume suitable data, if necessary.*

UNIT - I

- Q1)** a) Explain in details the performance characteristics of Pumps, Compressors and Blowers. **[10]**

OR

- Q2)** a) Explain the different applications of Compressors and Blowers. **[6]**
b) Define specific speed. What is its significance? **[4]**

UNIT - II

- Q3)** a) Explain with neat sketch working of a Reciprocating pump. **[4]**
b) A single acting Reciprocating pump has a plunger diameter of 30 cm and stroke of 20 cm. If the speed of the pump is 30 R.P.M. and it delivers 06.5 litres/sec of water, find the coefficient of discharge and the percentage slip of the pump. **[6]**

OR

- Q4)** a) Explain with neat sketch “Indicator Diagram of a Reciprocating Pump”. **[4]**
b) A double acting reciprocating pump has a stroke of 30 cm and a piston diameter of 15 cm. The delivery and suction heads are 26m and 04m. respectively, including friction heads. If the Pump is working at 60 R.P.M., find the Horse power required to drive the pump with 80% efficiency. **[6]**

P.T.O.

UNIT - III

- Q5)** a) Explain in details the “Design Procedure and Design Optimization of Pumps”. [6]
- b) Write a short note on “Selection of Pumps”. [4]

OR

- Q6)** a) Write a short note on “Selection of Materials for High Temperature and Corrosive Fluids”. [6]
- b) Write a short note on “Selection of Impellers”. [4]

